
Table of Contents

..... 1

```
% Lab 10: Comparison of Quadrature Rules
% Clenshaw-Curtis vs Gauss-Legendre

Nmax = 50;

% Exact integrals
Itrue = [ ...
    1/2;                                % I1 =  $\int |x|^3 dx$ 
    2*(exp(-1) + sqrt(pi)*(erf(1)-1)); % I2 =  $\int \exp(-x^{\{-2\}}) dx$ 
    pi/2;                               % I3 =  $\int 1/(1+x^2) dx$ 
    2/11                                % I4 =  $\int x^{\{10\}} dx$ 
];

labels = {'|x|^3', 'exp(-x^{\{-2\}})', '1/(1+x^2)', 'x^{\{10\}}'};

% Storage for errors
E_cc = zeros(4, Nmax);
E_ga = zeros(4, Nmax);

% Main Computation Loop
for N = 1:Nmax

    % Get nodes and weights
    [x_cc, w_cc] = clencurt(N);
    [x_ga, w_ga] = gauss(N);

    % Evaluate functions safely
    F_cc = test_functions(x_cc);
    F_ga = test_functions(x_ga);

    % Compute errors
    for k = 1:4
        E_cc(k,N) = abs(w_cc * F_cc(k,:) - Itrue(k));
        E_ga(k,N) = abs(w_ga * F_ga(k,:) - Itrue(k));
    end
end

% Plot: Clenshaw-Curtis Errors
plot_errors(E_cc, labels, Nmax, 'Clenshaw-Curtis Quadrature Errors');

% Plot: Gauss-Legendre Errors
plot_errors(E_ga, labels, Nmax, 'Gauss-Legendre Quadrature Errors');

% Degree of Exactness Test (Gauss-Legendre)
fprintf('\nDegree of Exactness Test for Gauss-Legendre:\n')
```

```

for N = 1:6
    [x,w] = gauss(N);
    approx = w * (x.^10);
    error = abs(approx - 2/11);

    fprintf('N = %d, Error for x^{10} = %.2e\n', N, error);
end

% Function: Test Functions
function F = test_functions(x)

    F = zeros(4, length(x));

    % (a) |x|^3
    F(1, :) = abs(x).^3;

    % (b) exp(-x^{-2}) (handle x=0 safely)
    xtmp = x;
    xtmp(xtmp==0) = eps;
    F(2, :) = exp(-xtmp.^(-2));

    % (c) 1/(1+x^2)
    F(3, :) = 1./(1 + x.^2);

    % (d) x^{10}
    F(4, :) = x.^10;
end

% Function: Plotting
function plot_errors(E, labels, Nmax, title_str)

    figure('Position',[100 100 900 900])
    sgtitle(title_str, 'FontSize', 18, 'FontWeight', 'bold')

    for k = 1:4
        subplot(2,2,k)
        semilogy(1:Nmax, E(k, :) + 1e-100, 'o-', 'LineWidth', 1.2)
        grid on

        xlim([1 Nmax])
        ylim([1e-18 1e+1])

        xlabel('N', 'FontSize', 14)
        ylabel('Error', 'FontSize', 14)
        title(labels{k}, 'FontSize', 15, 'FontWeight', 'bold')
    end
end

% Clenshaw-Curtis Quadrature
function [x,w] = clencurt(N)

    theta = pi*(0:N)'/N;
    x = cos(theta);

```

```

w = zeros(1,N+1);
ii = 2:N;
v = ones(N-1,1);

if mod(N,2)==0
    w(1) = 1/(N^2 - 1);
    w(N+1) = w(1);
    for k = 1:(N/2 - 1)
        v = v - 2*cos(2*k*theta(ii)) / (4*k^2 - 1);
    end
    v = v - cos(N*theta(ii))/(N^2 - 1);
else
    w(1) = 1/N^2;
    w(N+1) = w(1);
    for k = 1:((N-1)/2)
        v = v - 2*cos(2*k*theta(ii)) / (4*k^2 - 1);
    end
end

w(ii) = 2*v/N;
end

% Gauss-Legendre Quadrature
function [x,w] = gauss(N)

    beta = .5 ./ sqrt(1 - (2*(1:N-1)).^(-2));
    T = diag(beta,1) + diag(beta,-1);

    [V,D] = eig(T);
    x = diag(D);
    [x,ind] = sort(x);

    w = 2 * V(1,ind).^2;
end

```

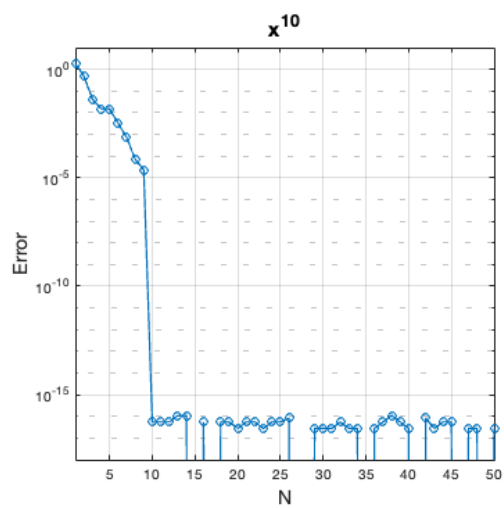
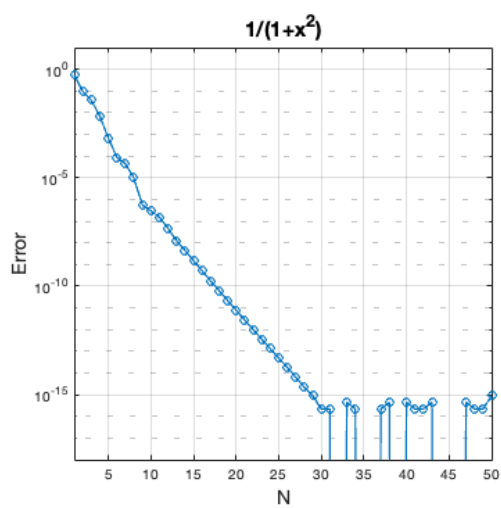
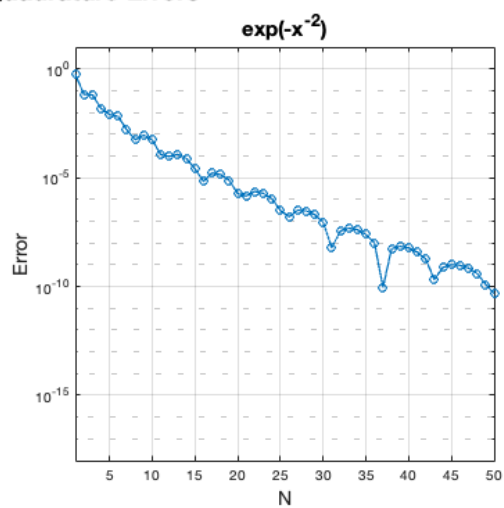
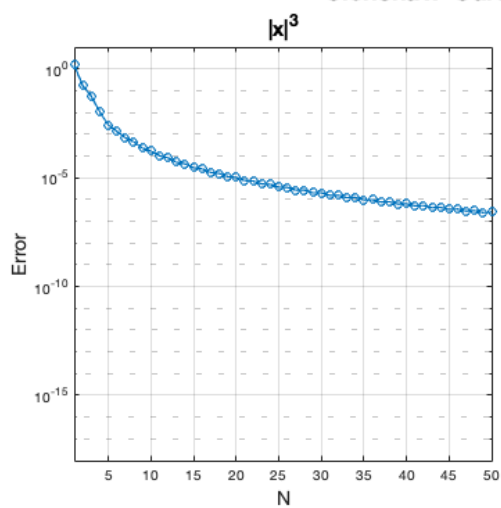
Degree of Exactness Test for Gauss-Legendre:

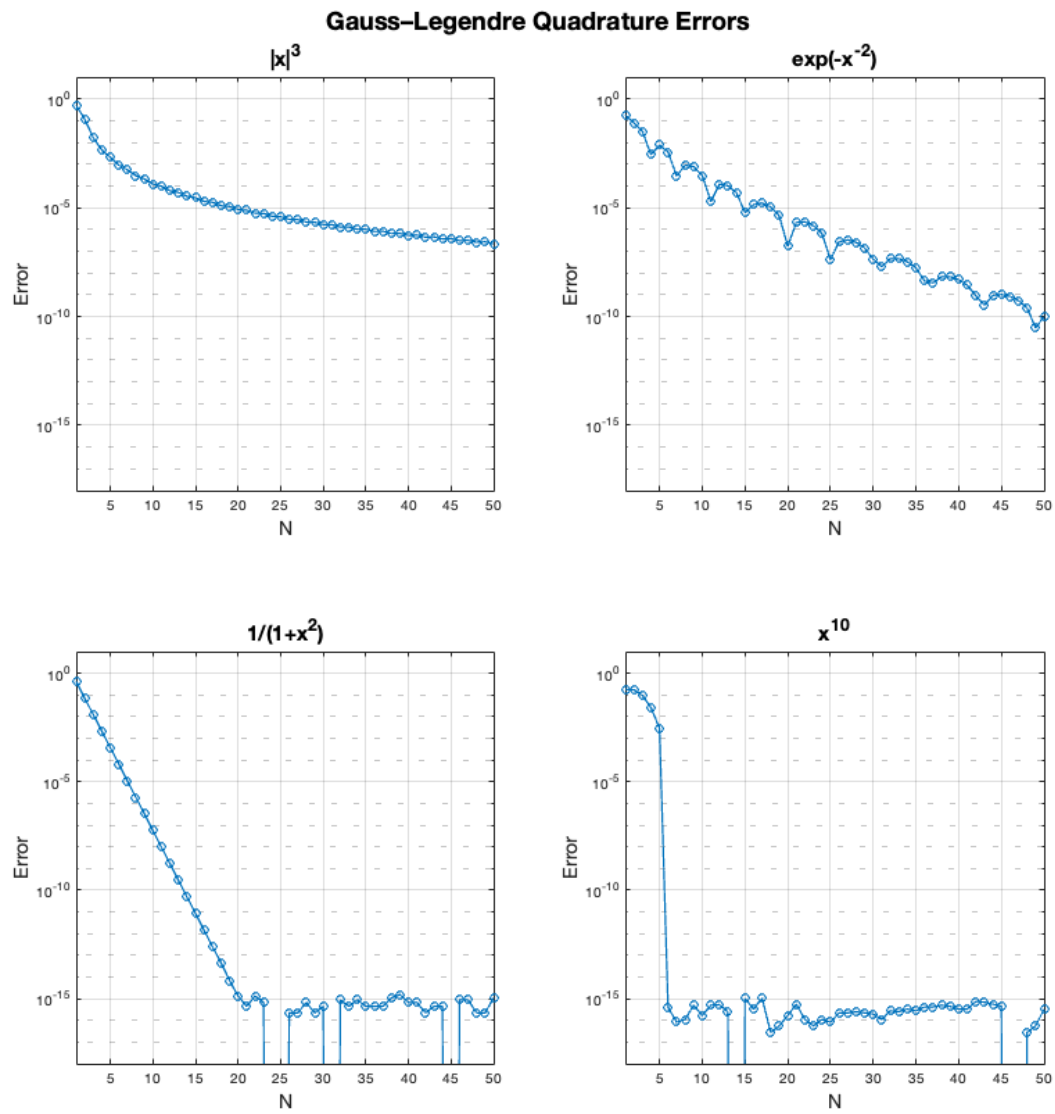
```

N = 1, Error for  $x^{10}$  = 1.82e-01
N = 2, Error for  $x^{10}$  = 1.74e-01
N = 3, Error for  $x^{10}$  = 9.54e-02
N = 4, Error for  $x^{10}$  = 2.58e-02
N = 5, Error for  $x^{10}$  = 2.93e-03
N = 6, Error for  $x^{10}$  = 3.89e-16

```

Clenshaw-Curtis Quadrature Errors





Published with MATLAB® R2024b